Succinct generators for a three-tensor torus action would imply NP is contained in P/poly

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Editorial note. Editable transcription only of formally published six-author CC BY3.0 Unported content; nonexclusive nativev2 is private aid and not a public original. CompleteSection3 author proof and bearingProposition21 included with necessaryDefinitions3/4 andSection2. Native NP-in-P/poly typo transcribed as the officially printed NP-subset-P/poly expression; no author mathematics repaired or extra assumptions added. Other original expressions remain as published. OmittedConjecture6 reference rendered as accurate external original location. Publisher packages replaced by minimalams/hyperref macros, no input/include/assets. All production/validation onlyYupeng; root owns compilation.

Editorial note. Revisionr002: split only the six-author title block into two lines via a begin-document author hook; the metadata authors array is unchanged.

Allowed standard prerequisites (editorial).

- Algebraic group actions/invariant rings, NP-completeness of3DM, efficient homogeneous components and rational-circuit PIT. Full formalSection2 standard statements plus Definitions3/4 retained; source-local matching-invariant characterization and reduction proof included completely

Group actions and invariant generators (original Section1)

An action (also called a representation) of a group $G \subseteq \mathrm{GL}_n(\mathbb{C})$ on an m-dimensional complex vector space V is a group homomorphism $\pi : G \to \mathrm{GL}_m(\mathbb{C})$, that is, an association of an invertible $m \times m$ matrix $\pi(g)$ for every group element $g \in G$, satisfying $\pi(g_1 g_2) = \pi(g_1)\pi(g_2)$ for all $g_1, g_2 \in G$ (and $\pi(e) = I_m$, where $e \in G$ is the identity element and I_m is the identity matrix). A group element $g \in G$ acts on a vector $v \in V$ by the linear transformation $\pi(g)$, and in this paper we will be dealing with algebraic actions, that is, the entries of the matrix $\pi(g)$ will be rational functions in the entries of the matrix g. We will write $g \cdot v = \pi(g)v$. We denote by $\mathbb{C}[V]$ the ring of polynomial functions on V. A polynomial function $f \in \mathbb{C}[V]$ is called invariant if $f(g \cdot v) = f(v)$ for all

$v \in V$ and $g \in G$. The induced action on polynomials is $(g \cdot p)(v) = p(g^{-1} \cdot v)$. A set $\{f_i\}_{i \in I}$ of invariant polynomials is called a generating set if any other invariant polynomial can be written as a polynomial in the f_i's. The invariant ring is denoted $\mathbb{C}[V]^G$.

Definition 3 (**Succinct encoding of generators**)**.** *Fix an action of a group G on a vector space $V = \mathbb{C}^m$. We say that an arithmetic circuit $\mathcal{C}(x_1, \ldots, x_m, y_1, \ldots, y_r)$* succinctly encodes *the generators of the invariant ring if the set of polynomials formed by evaluating the y-variables, $\{\mathcal{C}(x_1, \ldots, x_m, \alpha_1, \ldots, \alpha_r)\}_{\alpha_1, \ldots, \alpha_r \in \mathbb{C}}$, is a generating set for the invariant ring $\mathbb{C}[V]^G$.*

Remark 4. *The* size *of a succinct encoding as defined above is given by the size of the circuit $\mathcal{C}(x_1, \ldots, x_m, y_1, \ldots, y_r)$, which is measured by the bit complexity of the constants used in the computation of $\mathcal{C}$ as well as the number of gates of the computation graph of $\mathcal{C}$. In particular, this means that all constants used in the computation of $\mathcal{C}$ are rationals.*

For our first counterexample, we analyze a simple (torus) action on 3-tensors. Here, $\mathrm{ST}_n(\mathbb{C})$ denotes the group of $n \times n$ diagonal matrices with determinant 1.

Theorem 8. *Consider the natural action of $G = \mathrm{ST}_n(\mathbb{C}) \times \mathrm{ST}_n(\mathbb{C}) \times \mathrm{ST}_n(\mathbb{C})$ on $V = \mathbb{C}^n \otimes \mathbb{C}^n \otimes \mathbb{C}^n$. Then any set of generators for the invariant ring cannot have a polynomial sized (in n) succinct encoding, unless* $\mathrm{NP} \subseteq \mathrm{P/poly}$.

2 Preliminaries

In this section we establish notation and we formally state basic facts and definitions which we will need in later sections.

Definition 16 (3-dimensional matching [4])**.** *The* 3-dimensional matching *problem is defined as follows:*

Input: *a set $U \subseteq [n] \times [n] \times [n]$, representing the edges of a tripartite, 3-uniform hypergraph.*

Output: *YES, if there is a set of hyperedges $W \subseteq U$ such that $|W| = n$ and no two elements of W agree in any coordinate (that is, they form a matching in this hypergraph). NO, if there is no such set.*

Theorem 17 (NP-completeness of 3-dimensional matching [4])**.** *The 3-dimensional matching problem is NP-complete.*

2.1 Basic facts from algebraic complexity

We now give basic facts that from algebraic complexity which we will use in the next sections.

The next proposition shows that homogeneous components of low degree of an arithmetic circuit can be efficiently computed, with a small blow-up in circuit size and without the use of any extra constants. This proposition was originally proved by Strassen in [6] and its proof can be found in [7, Theorem 2.2]. In the following proposition, given a polynomial $p(\mathbf{x})$, we denote its degree-d homogeneous component by $H_d[p(\mathbf{x})]$.

Proposition 18 (Efficient computation of homogeneous components)**.** *Given a circuit $\mathcal{C}(\boldsymbol{x})$ of size s, then for every $r \in \mathbb{N}$ there is a homogeneous circuit $\Psi(\boldsymbol{x})$ of size $O(r^2 s)$ computing $H_0[\mathcal{C}(\boldsymbol{x})], H_1[\mathcal{C}(\boldsymbol{x})], \ldots, H_r[\mathcal{C}(\boldsymbol{x})]$. Moreover, the constants used in the computation of the components $H_i[\mathcal{C}(\boldsymbol{x})]$ are a subset of the coefficients used in the computation of $\mathcal{C}(\boldsymbol{x})$.*

The next theorem, proved by [1, Theorem 4.10] gives us a randomized polynomial time algorithm to test whether an algebraic circuit of polynomial size, with rational coefficients, is identically zero. Another randomized algorithm easily follows from [5, Lemma 2], when adapted for polynomials with rational coefficients.

Theorem 19 (PIT for poly-sized circuits [1])**.** *Let $P(\boldsymbol{x}) \in \mathbb{Q}[\boldsymbol{x}]$ be a polynomial in the variables $\boldsymbol{x} = (x_1, \ldots, x_n)$, with each variable x_i having degree bounded by d_i, and whose coefficients are rationals with bit complexity bounded by B. If $P(\boldsymbol{x})$ is given as an arithmetic circuit of size s, then there exists a randomized algorithm running in time $\mathrm{poly}(n, s, \log(B), 1/\epsilon)$ and using $O\left(\sum_{i=1}^{n} \log(d_i) + \log(B)\right)$ random bits which tests whether $P(\boldsymbol{x})$ is identically zero. If $P(\boldsymbol{x})$ is the identically zero polynomial then the algorithm always succeeds. Otherwise, it errs with probability at most ϵ.*

3 Hardness of Generators for torus actions

Let $\mathbb{C}^*$ denote the multiplicative group consisting of all non-zero complex numbers. A direct product $\mathrm{T}_n = (\mathbb{C}^*)^n$ is called a torus, and is clearly an abelian group. Tori are important examples of reductive groups – any abelian connected reductive group is a torus! It is often the case that it is easier to understand tori in comparison with more general (non-abelian) reductive groups. This is no different for invariant theory, see for example [2, 8]. We also point to [3, Proposition 3.3] for an elementary linear algebraic description of the invariant ring for torus actions. Conjecture 6 of the original publication already fails in this well behaved setting. This is the content of our Theorem 8, which we will prove in this section. Recall that $\mathrm{ST}_n(\mathbb{C}) \cong \{\mathbf{z} \in \mathrm{T}_n : z_1 \cdots z_n = 1\}$, which is itself a torus.

Theorem 20 (Theorem 8, restated)**.** *Consider the natural action of $G = \mathrm{ST}_n(\mathbb{C}) \times \mathrm{ST}_n(\mathbb{C}) \times \mathrm{ST}_n(\mathbb{C})$ on $V = \mathbb{C}^n \otimes \mathbb{C}^n \otimes \mathbb{C}^n$, where an element $(\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}) \in G$ acts on a tensor $u \in V$ as follows: $(\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}) \cdot u := v$, such that $v_{ijk} = a_i b_j c_k u_{ijk}$. Any set of generators for the invariant ring of this action cannot have a polynomial sized (in n) succinct encoding, unless $\mathrm{NP} \subseteq \mathrm{P/poly}$.*

Proof. Suppose that the natural action above has a set of generators with a polynomial sized succinct encoding. Thus, there is an arithmetic circuit $\mathcal{C}(\mathbf{x}, \mathbf{y})$ of size $s = \mathrm{poly}(n)$, where $\mathbf{x} = (x_{ijk})_{i,j,k=1}^{n}$ is the set of variables corresponding to V and $\mathbf{y} = (y_1, \ldots, y_r)$ is the set of auxiliary variables, with $r = \mathrm{poly}(n)$. Moreover, from the definition of the size of a succinct encoding we also have that the constants used in the computation of $\mathcal{C}(\mathbf{x}, \mathbf{y})$ are rational numbers with bit complexity bounded by $b = \mathrm{poly}(n)$. In particular, $\mathcal{C}(\mathbf{x}, \mathbf{y}) \in \mathbb{Q}[\mathbf{x}, \mathbf{y}]$.

Let us consider the circuit $\mathcal{C}(\mathbf{x}, \mathbf{y})$ as a circuit whose constants are in $\mathbb{Q}[\mathbf{y}]$ and whose variables are only the $\mathbf{x}$ variables, that is, a circuit in $\mathbb{Q}[\mathbf{y}][\mathbf{x}]$. Then, Proposition 18 tells us that there exists a homogeneous circuit $\mathcal{C}_n(\mathbf{x}, \mathbf{y})$, in the

$\mathbf{x}$ variables, of degree n and size $O(n^2 s)$ that computes the homogeneous component of $\mathcal{C}(\mathbf{x}, \mathbf{y})$ of degree n as a function of $\mathbf{x}$. Moreover, the constants of this circuit are a subset of the constants used in the circuit $\mathcal{C}(\mathbf{x}, \mathbf{y})$. Since we consider the latter as a circuit in only the $\mathbf{x}$ variables, the constants in this case are given by the elements of $\mathbb{Q}$ used in the computation of $\mathcal{C}$ as well as the auxiliary variables $\mathbf{y}$. In particular, $\mathcal{C}_n(\mathbf{x}, \mathbf{y})$ can be written in the following way:

$$\mathcal{C}_n(\mathbf{x}, \mathbf{y}) = \sum_{\mathbf{m} \in \mathcal{N}_n(\mathbf{x})} f_{\mathbf{m}}(\mathbf{y}) \cdot \mathbf{m}, \tag{1}$$

where $\mathcal{N}_n(\mathbf{x})$ is the set of all monomials of degree n in the variables $\mathbf{x}$ and $f_{\mathbf{m}}(\mathbf{y})$ are polynomials in the variables $\mathbf{y}$ of degree at most 2^s, as the circuit $\mathcal{C}$ has size at most s.

In Proposition 21 below, we will show that the invariants of minimum degree of our action are in degree n, and these are spanned by the (maximum) 3-dimensional matching monomials. Thus, if a monomial of degree n is invariant under our action, it must be the case that this monomial corresponds to a 3-dimensional matching. Moreover, the action maps any monomial (invariant or not) to a constant times itself. As $\mathcal{C}_n(\mathbf{x}, \mathbf{y})$ must only compute invariant polynomials, this implies that equation (1) is actually of the following form:

$$\mathcal{C}_n(\mathbf{x}, \mathbf{y}) = \sum_{\mathbf{m} \in \mathcal{M}_n(\mathbf{x})} f_{\mathbf{m}}(\mathbf{y}) \cdot \mathbf{m}, \tag{2}$$

where $\mathcal{M}_n(\mathbf{x})$ is the set of all 3-dimensional matching monomials in the variables $\mathbf{x}$. Moverover, since $\mathcal{C}(\mathbf{x}, \mathbf{y})$ succinctly encodes of a set of generators, the span of $\{\mathcal{C}_n(\mathbf{x}, \alpha)\}_{\alpha \in \mathbb{C}^r}$ must necessarily be the same as the span of the 3-dimensional matching polynomials.

We will now show that the existence of the circuit $\mathcal{C}_n(\mathbf{x}, \mathbf{y})$ implies that NP $\subseteq$ P/poly. For that purpose, we will show that given $\mathcal{C}_n(\mathbf{x}, \mathbf{y})$ one can solve the 3-dimensional matching problem in P/poly. Let H be a tripartite 3-uniform hypergraph, whose edges are given by a subset $E \subseteq [n] \times [n] \times [n]$. We can associate to this graph the tensor $v \in V$ where $v_{ijk} = 1$ if hyperedge $(i, j, k) \in E$ and $v_{ijk} = 0$ otherwise. Note that H has a 3-dimensional matching of size n if and only if at least one of the 3-dimensional matching monomials *does not vanish* on our tensor v. This last condition is equivalent to the fact that the circuit $\mathcal{C}_n(v, \mathbf{y})$ does not compute the zero polynomial (as we know that the span of the set $\{\mathcal{C}_n(\mathbf{x}, \alpha)\}_{\alpha \in \mathbb{C}^r}$ is the same as the span of all 3-dimensional matching monomials). Thus, to solve the 3-dimensional matching problem in P/poly it is enough to give a randomized polynomial time algorithm for testing whether $\mathcal{C}_n(v, \mathbf{y})$ is the zero polynomial or not.[1]

Since $\mathcal{C}_n(v, \mathbf{y})$ is a circuit of size poly(n) with rational constants of bit complexity poly(n), it computes a polynomial $P(\mathbf{y})$ with rational coefficients having bit complexity at most exp(poly(n)) and degree at most exp(poly(n)). This is the setting in which Theorem 19 applies, giving us the desired randomized polynomial time algorithm. This concludes our proof modulo Proposition 21, which we will now turn our attention to. □

In the following proposition, we denote by $\mathcal{S}_n$ the symmetric group on n letters.

[1] It is enough to give a randomized polynomial time algorithm because we know that BPP/poly = P/poly.

Proposition 21. *The maximum 3-dimensional matching monomials* $\prod_{i=1}^{n} x_{i\sigma(i)\tau(i)}$*, where* $\sigma, \tau \in \mathcal{S}_n$*, span the invariants of degree* n *of the natural action of* $G = \mathrm{ST}_n(\mathbb{C}) \times \mathrm{ST}_n(\mathbb{C}) \times \mathrm{ST}_n(\mathbb{C})$ *on* $V = \mathbb{C}^n \otimes \mathbb{C}^n \otimes \mathbb{C}^n$*. Moreover, there are no nonconstant invariants of degree less than* n *for this action.*

Proof. Since the action maps any monomial to a constant times itself, it is easy to see that the invariant polynomials are generated by invariant monomials. To prove the proposition, it is therefore enough to show that the matching monomials are invariant, that there are no other invariant monomials of degree n, and that there are no invariant monomials of smaller degree.

We first prove that the matching monomials are invariant. Note that the natural action of G on V induces the following action on the variables x_{ijk}: $(\mathbf{a}, \mathbf{b}, \mathbf{c}) \cdot x_{ijk} = (a_i b_j c_k)^{-1} \cdot x_{ijk}$. [2] Additionally, note that $\prod_{\ell=1}^{n} a_\ell = \prod_{\ell=1}^{n} b_\ell = \prod_{\ell=1}^{n} c_\ell = 1$. Given a matching monomial $\prod_{i=1}^{n} x_{i\sigma(i)\tau(i)}$, we therefore have that

$$\begin{aligned}(\mathbf{a}, \mathbf{b}, \mathbf{c}) \cdot \prod_{i=1}^{n} x_{i\sigma(i)\tau(i)} &= \prod_{i=1}^{n} \left((a_i b_{\sigma(i)} c_{\tau(i))})^{-1} \cdot x_{i\sigma(i)\tau(i)} \right) \\ &= \prod_{i=1}^{n} (a_i b_{\sigma(i)} c_{\tau(i)})^{-1} \cdot \prod_{i=1}^{n} x_{i\sigma(i)\tau(i)} \\ &= \prod_{i=1}^{n} x_{i\sigma(i)\tau(i)}\end{aligned}$$

where in the last equality we note that for any permutation $\sigma \in \mathcal{S}_n$ (or τ) we have $1 = \prod_{\ell=1}^{n} a_\ell = \prod_{\ell=1}^{n} a_{\sigma(\ell)}$ (and similarly for $\mathbf{b}$ and $\mathbf{c}$). This proves that the matching monomials are invariant monomials of the natural G-action on V.

Now, let us prove that no other monomial of degree n is an invariant for this action. Let $\prod_{m=1}^{n} x_{i_m j_m k_m}$ be a monomial, where $(i_m, j_m, k_m) \in [n]^3$, that is not a matching monomial. Then there exists some coordinate, w.l.o.g. the first coordinate, for which the set $\{i_m\}_{m=1}^{n}$ is a strict subset of $[n]$. Equivalently, there is an element $\ell \in [n]$ such that $\ell \notin \{i_m\}_{m=1}^{n}$. W.l.o.g., we can assume that $\ell = 1$. Thus, the action of $\mathbf{a} = (\alpha^{n-1}, \alpha^{-1}, \ldots, \alpha^{-1}), \mathbf{b} = \mathbf{c} = (1, \ldots, 1)$ on our monomial $\prod_{m=1}^{n} x_{i_m j_m k_m}$ is as follows:

$$(\mathbf{a}, \mathbf{b}, \mathbf{c}) \cdot \prod_{m=1}^{n} x_{i_m j_m k_m} = \alpha^n \cdot \prod_{m=1}^{n} x_{i_m j_m k_m}$$

which proves that this monomial is not an invariant. This completes the proof that the matching monomials span the invariants of degree n.

Now we are left with proving that there are no nonconstant monomials of degree less than n that are invariant. Note that if we have a monomial with degree less than n, we can represent it as $\prod_{m=1}^{d} x_{i_m j_m k_m}$, where $d < n$ and by the pigeonhole principle, we know that there exists $\ell \in [n]$ such that ℓ does not appear as a first coordinate entry in the set of tuples $\{(i_m, j_m, k_m)\}$. If $d > 0$ then, analogously to the previous paragraph, we know that such monomials cannot be invariants of the natural action of G over V, therefore showing that no nonconstant monomial of degree $< n$ can be an invariant. This completes the proof. □

[2] The inverse comes from the general formula $(g \cdot p)(v) := p(g^{-1} \cdot v)$.

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